Introduction to GPU Programming

Linux CLI GPU Device Identification

nvidia-smi CLI Basic Use

nvidia-smi

Executing without options will give a minimum of the following information:

- The number of Nvidia GPUs
- Product information for each Nvidia GPU, which includes the following:
  - Product name
  - Utilization stats for GPU and its memory
  - Temperature
  - Clock speed for graphics interface, streaming multiprocessors, and memory
  - Memory stats for total, used, and free

[1]

nvidia-smi CLI Advanced Use

CLI options:

- -L, --list_gpus – This will only include the list of GPUs and their UUIDs.
- -i, --id=ID – this will filter down the GPUs listed to only the one with this UUID
- -d TYPE, --display=TYPE – this will filter down the GPU data that is listed to TYPE. Examples of TYPE are MEMORY, UTILIZATION, and CLOCK

[1]

lspci CLI

- Not installed on all Nvidia CUDA-enabled machines, so don’t presume that you have access to the command
- lspci lists all PCI devices connected to the current machine
- -v option of more verbose output (note can also do –vv and –vvv for more verbosity)
- lspci | grep NVIDIA will show Nvidia devices
- lspci | grep -i --color 'vga\|3d\|2d’ for more colorful highlighting of VGA-based PCI devices, which will include Nvidia GPUs

Bibliography

- [1] 2016. Nvidia. nvidia-smi documentation. Retrieved from http://developer.download.nvidia.com/compute/DCGM/docs/nvidia-smi-367.38.pdf.

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